

Prerequisite Diagnostic Quiz

M 375T Mathematical Foundations of Machine Learning — Fall 2026

15 minutes · closed book · no calculators · show your work in the boxes

Codename (as in the survey): _____

*This quiz checks the floor this course stands on: if it feels far out of reach, come talk to me this week (office hours Tue/Thu 5–6 PM, PMA 9.160) — the last day to drop without permission is **Wednesday, September 9**.*

Linear algebra.

1. Let $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ and $v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Compute Av . [2 pts]

Answer:

2. Solve the system $\begin{cases} x + 2y = 5 \\ 3x + 4y = 11 \end{cases}$ [2 pts]

Answer:

3. Let $M = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$. Verify that $v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector of M , and find its eigenvalue. [2 pts]

Answer:

Probability.

4. You roll two fair six-sided dice. What is $\mathbb{P}(\text{sum} = 12)$? What is $\mathbb{P}(\text{sum} \geq 11)$? [2 pts]

Answer:

5. A random variable X takes the value -1 with probability $\frac{1}{2}$, the value 1 with probability $\frac{1}{4}$, and the value 3 with probability $\frac{1}{4}$. Compute $\mathbb{E}[X]$ and $\text{Var}(X)$. [2 pts]

Answer:

6. Let $Z \sim N(0, 1)$ be a standard Gaussian, with density $\varphi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$.
- (a) Write $\mathbb{P}(Z > 0)$ explicitly as an integral of φ , then evaluate it (no computation needed — think symmetry).
 - (b) Let $X = 2Z + 1$. What are the mean and the variance of X ?

[2 pts]

Answer:

Total: 12 points. 10–12: ready. 8–9: ready with review (Calder–Olver Ch. 1–3, §7.1). ≤ 7 : come see me this week.